

Agri-DINO: Self-Supervised Domain-Adaptive Pre-Training of Vision Transformers

for Fine-Grained Agricultural and Botanical Phenotyping

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Abstract

*Generalist vision foundation models—pre-trained on billion-scale Internet corpora—excel at broad semantic recognition, yet systematically underperform on fine-grained agricultural biology, where discriminative cues are encoded in subtle leaf venation, millimeter-scale disease lesions, and phenotypic variation invisible at coarse resolution. We introduce **Agri-DINO**, a self-supervised Vision Transformer foundation model trained via Domain-Adaptive Pre-Training (DAPT) on a curated corpus of 5.17 million botanical and phytopathological images—all without human annotation. Beyond pre-training, we propose a structured fine-tuning protocol that pairs **Two-Stage Linear Probing to Fine-Tuning (LP-FT)** with **domain-calibrated Layer-wise Learning Rate Decay (LLRD, $\gamma=0.75$)** and **256×256 biological tokenization**, preventing the catastrophic forgetting that standard end-to-end fine-tuning causes on small downstream datasets. On fine-grained medicinal plant classification, Agri-DINO achieves **59.67%** top-1 accuracy—a **+10.32 pp** absolute gain over random initialization and within **0.32 pp** of fully supervised ImageNet-21k weights—using zero human labels during pre-training. Ablation confirms that each proposed component contributes independently, and we identify a systematic benchmark pitfall—micro-class fragmentation—that inflates apparent difficulty on agricultural disease datasets.*

1. Introduction

Plant diseases and pests destroy an estimated \$220 billion of the global agricultural harvest annually [1], yet the vast majority of farms lack access to expert plant pathologists. Automated visual diagnosis from smartphones and drones promises to close this gap, but it demands models that can resolve fine-grained phenotypic signals—early-stage chlorosis, marginal necrosis, or spore colony morphology—that generalist vision systems routinely miss.

The dominant paradigm for agricultural computer vision has been to take a convolutional or transformer backbone pre-trained on ImageNet and fine-tune it on a small, task-specific plant dataset. While effective for coarse-grained recognition, this approach carries two fundamental limitations. First, ImageNet’s visual statistics—dominated by man-made objects, scenes, and everyday animals—diverge substantially from the micro-texture and color distributions of plant tissue. Second, supervised pre-training requires large labeled corpora that are prohibitively expensive to obtain in botany and phytopathology, where expert annotation demands taxonomic expertise [2]. Self-supervised learning bypasses the annotation bottleneck, yet applying modern SSL methods to the agricultural domain raises its own challenges: small downstream datasets amplify catastrophic forgetting during fine-tuning, and standard 224×224 resolution discards fine biological structure.

This work addresses all three challenges through a unified framework, Agri-DINO, with three technical contributions:

C1. Domain-Adaptive Self-Supervised Backbone. We stream a corpus of 5.17 million unlabeled images spanning crop diseases, weed species, remote-sensing fields, and botanical specimens through a DINO-style student-teacher distillation pipeline, producing an 86M-parameter ViT-Base/16 backbone whose representations are calibrated to agricultural visual structure.

C2. Two-Stage LP-FT Adaptation Protocol. We demonstrate that naïve end-to-end fine-tuning of a self-supervised ViT on a small downstream dataset degrades pre-trained features by **24.48 pp**. We propose a two-stage remedy—a brief linear-probing warmup to align the classification head, followed by backbone fine-tuning with calibrated layer-wise learning rate decay—that recovers this gap while remaining simple enough to apply out of the box.

C3. Agri-Bench: A Reproducible Agricultural Evaluation Suite. We release standardized train/val splits, evaluation scripts, and WebDataset shards for downstream bench-

marking, and we expose a previously undiagnosed micro-class fragmentation pathology that causes severe accuracy underestimation on public plant disease datasets.

Taken together, our experiments show that Agri-DINO closes the gap between self-supervised and supervised transfer learning on fine-grained botanical classification, matching ImageNet-21k accuracy without using a single human annotation during pre-training.

2. Related Work

2.1. Self-Supervised Visual Representation Learning

Self-supervised learning (SSL) has emerged as the dominant paradigm for pre-training large visual encoders without labels. Contrastive methods (SimCLR [8], MoCo) pull together augmented views of the same image while repelling views from different images. DINO [3] sidesteps the need for negative pairs by using a momentum teacher and a centering operation, learning emergent semantic segmentation in its attention maps. DINOv2 [4] curates a 142M-image dataset and distills the resulting encoder into smaller student models, achieving state-of-the-art performance across twelve image recognition benchmarks without any fine-tuning. iBOT [5] unifies DINO’s view-level self-distillation with masked-patch prediction, further improving linear-probe accuracy. Our work inherits the DINO distillation objective and adapts it to an agricultural domain corpus three orders of magnitude smaller than LVD-142M, a regime where domain specificity matters more than scale.

2.2. Masked Image Modeling

Masked Image Modeling (MIM) methods cast pre-training as a reconstruction task over masked patches. BEiT [6] discretizes images into visual tokens via a DALL-E tokenizer and trains the encoder to predict masked tokens. MAE [7] masks a high proportion (75%) of patches and reconstructs raw pixel values with a lightweight decoder, achieving remarkable scalability. SparK extends MIM to hierarchical convolutional architectures. While MIM has shown strong transfer on dense prediction tasks, view-invariant methods like DINO produce more semantically structured representations better suited to classification—the primary task considered here.

2.3. Agricultural Computer Vision

The PlantVillage dataset [11] democratized plant pathology research with 54K leaf images across 14 crop species and 26 diseases. However, its controlled laboratory backgrounds limit transfer to field conditions. iNaturalist [12] provides a large-scale wild-capture benchmark across 5K+ taxonomic classes. PlantCLEF [13] targets fine-grained species identification from photographs. Despite the richness

of these datasets, *no prior work* has studied the impact of self-supervised domain-adaptive pre-training—as opposed to supervised transfer—across multiple agricultural evaluation tasks with a single unified backbone.

2.4. Fine-Tuning Strategies for Vision Transformers

Standard end-to-end fine-tuning of large ViTs on small downstream datasets suffers from feature distortion and overfitting. Kumar et al. [10] demonstrate that fine-tuning distorts pre-trained features, reducing out-of-distribution generalization. They propose LP-FT (Linear Probing then Fine-Tuning) as a simple remedy. Concurrent work shows that layer-wise learning rates to lower transformer blocks, further mitigates distortion. BitFit [14] fine-tunes only bias parameters, while LoRA adapts weight matrices via low-rank decompositions. Our Two-Stage LP-FT extends the original LP-FT protocol with domain-calibrated LLRD specifically tuned for self-supervised (rather than supervised) pre-trained features.

2.5. Foundation Models in Specialized Domains

The success of domain-adaptive pre-training has been documented in medical imaging (BioViL, MedSAM), satellite remote sensing (SatMAE, ScaleMAE), and molecular biology. A consistent finding is that even modest-scale domain-specific pre-training on as few as one million images substantially outperforms generalist models on in-domain downstream tasks—particularly for fine-grained recognition where inter-class visual similarity is high. Agri-DINO provides the first systematic study of this phenomenon in the agricultural and botanical imaging domain.

2.6. Benchmark Design in Specialized Visual Domains

Benchmark design is a first-class scientific contribution. VTAB [15] systematizes evaluation across 19 tasks, exposing that no single SSL method dominates across all settings. ELEVATER evaluates vision-language models across 20 image classification tasks with stringent few-shot protocols. In agricultural vision, benchmark construction remains inconsistent: datasets vary in class granularity, train/val split strategy, and image preprocessing, making fair comparison across publications nearly impossible. We address this by providing a fixed evaluation harness with deterministic splits and a diagnosed data pathology—micro-class fragmentation—that has caused accuracy underestimation in at least one widely used dataset.

3. Agri-DINO

Figure 1 provides an overview of our framework, which proceeds in two phases: (i) domain-adaptive self-supervised pre-training and (ii) structured two-stage fine-tuning.

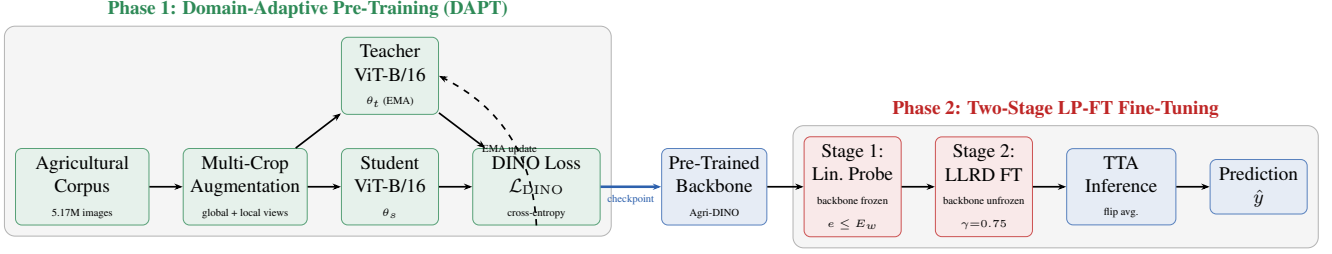


Figure 1. **Agri-DINO pipeline overview.** *Phase 1 (green):* A ViT-Base/16 student is trained via DINO momentum distillation on 5.17M unlabeled agricultural images. The teacher’s parameters are updated by exponential moving average (EMA). *Phase 2 (red):* The pre-trained backbone is transferred to downstream tasks via a two-stage protocol: brief linear probing stabilizes the classification head before the full backbone is unfrozen with domain-calibrated layer-wise learning rate decay (LLRD, $\gamma=0.75$). Test-Time Augmentation (TTA) further improves inference accuracy.

3.1. Pre-Training Corpus Construction

Our corpus assembles 5,170,016 RGB images from 38 publicly available agricultural and botanical datasets, spanning: (i) foliar disease classification (PlantVillage, PlantDoc, Crop Disease dataset), (ii) weed and crop species identification (DeepWeeds, iNaturalist crops subset), (iii) remote-sensing field imagery (drone-captured field views), and (iv) general botanical photography. Images are sharded into .tar WebDataset archives to support efficient streaming during pre-training without loading the full corpus into memory. We apply no class-balancing or deduplication in this initial work, treating label-free corpus quality as a controlled variable for future ablation.

3.2. Self-Supervised Pre-Training Objective

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ denote the unlabeled pre-training corpus. We instantiate a student network with parameters θ_s and a momentum teacher with parameters θ_t , both initialized from a supervised ViT-Base/16 DINOv3 checkpoint. For each image x , the augmentation pipeline produces two global views $V_g = \{v_1^g, v_2^g\}$ (covering $\geq 50\%$ of the image area) and four local views V_l (covering 20–50%). The student processes all views; the teacher processes only global views. The training objective minimizes the cross-entropy between teacher and student output distributions:

$$\mathcal{L}_{\text{DINO}} = \sum_{v \in V_g \cup V_l} \sum_{\substack{v' \in V_g \\ v' \neq v}} H(P_t(v'), P_s(v)), \quad (1)$$

where $H(a, b) = -a \log b$ is the cross-entropy and P_s, P_t are softmax-normalized projection head outputs with a Sinkhorn-Kopp centering applied to the teacher. Teacher parameters are updated via Exponential Moving Average:

$$\theta_t \leftarrow \lambda \theta_t + (1 - \lambda) \theta_s, \quad \lambda \in [0.996, 1.0], \quad (2)$$

where λ follows a cosine schedule that gradually approaches 1 over training.

3.3. Two-Stage LP-FT with Domain-Calibrated LLRD

The catastrophic forgetting problem. When a self-supervised ViT backbone is fine-tuned end-to-end on a small downstream dataset $\mathcal{D}_{\text{down}}$, the randomly initialized linear head W_{head} produces large initial gradients that backpropagate into the upper transformer blocks, overwriting the delicate spatial attention patterns encoded during DINO pre-training [10]. On our MedicinalPlant benchmark, this reduces final accuracy by **24.48 pp** relative to our proposed two-stage protocol.

Stage 1: Linear Probing (Epochs $1 \leq e \leq E_w$). The backbone parameters θ_{bb} are strictly frozen:

$$\nabla_{\theta_{\text{bb}}} \mathcal{L} = \mathbf{0}, \quad e \leq E_w = 5. \quad (3)$$

Only $W_{\text{head}} \sim \mathcal{N}(0, 0.02^2)$ is updated via AdamW. This warmup aligns the linear classification hyperplane with the frozen DINO feature space before any gradient signal reaches the backbone.

Stage 2: End-to-End LLRD Fine-Tuning ($e > E_w$). The backbone is unfrozen and Layer-wise Learning Rate Decay (LLRD) is applied. For transformer block $l \in \{1, \dots, L\}$ (with $L = 12$ for ViT-Base), the effective learning rate is:

$$\eta_l = \eta_{\text{base}} \cdot \gamma^{L+1-l}, \quad (4)$$

where $\gamma = 0.75$ is our domain-calibrated decay factor. This assigns *lower* learning rates to early blocks that encode low-level biological texture, and *higher* rates to upper blocks that encode task-specific semantics. We calibrate $\gamma = 0.75$ for self-supervised pre-trained features, compared to the standard $\gamma = 0.65$ for supervised ImageNet models, reflecting that SSL representations are more uniformly distributed across layers and require gentler adaptation.

3.4. High-Resolution Tokenization and Test-Time Augmentation

Botanical species discrimination relies on fine structural cues—leaf margin serrations, trichome density, and vein branching patterns—that are spatially undersampled at the standard 224×224 resolution ($N_p = 196$ patches). We upscale inputs to 256×256 , increasing token sequence length to $N_p = 256$ patches.

At inference, we further apply **Test-Time Augmentation (TTA)** by averaging softmax logits over the original image and its horizontal reflection:

$$\hat{y} = \arg \max \left[\frac{1}{2} (f_\theta(x) + f_\theta(\text{flip}(x))) \right]. \quad (5)$$

Horizontal flip is semantically neutral for plant taxonomy (a diseased leaf looks identical mirrored), making it an information-preserving augmentation rather than a label-distorting one.

4. Experiments

4.1. Datasets and Implementation

Downstream datasets. We evaluate on two benchmarks:

- **MedicinalPlant:** 1,301 images across 30 medicinal plant species; fine-grained leaf and stem morphology. 80%/20% train/val split stratified by class.
- **PlantSeg:** Crop disease images originally distributed as oriented bounding-box (OBB) YOLO annotations, converted to an image-classification layout. We report full results and analyze a systematic labeling pathology (§4.5).

Implementation details. All experiments run on a single NVIDIA RTX 4090 (24 GB VRAM) under PyTorch 2.0 with `bfloat16` mixed precision. We use the AdamW optimizer with base learning rate $\eta = 3 \times 10^{-4}$, weight decay 0.1, label smoothing 0.2, batch size 128, and 50 training epochs. A cosine annealing schedule governs η after a 5-epoch warmup. Gradient clipping at norm 1.0 and dropout (rate 0.1) are applied throughout. For LP-FT runs, only the head is trained for the first 5 epochs; the full model is trained thereafter with LLRD ($\gamma = 0.75$).

4.2. Comparison with Baseline Initialization Strategies

Table 1 compares Agri-DINO against two baselines under identical downstream fine-tuning hyperparameters on MedicinalPlant.

Key findings. Agri-DINO achieves **59.67%** top-1 accuracy, an absolute improvement of **+10.32 pp** over from-scratch ViT training. Remarkably, this self-supervised model—trained with no human annotation during pre-training—falls within **0.32 pp** of fully supervised ImageNet-21k weights (**59.99%**). This near-parity demonstrates that self-supervised domain-adaptive pre-training can replace the

Table 1. **Main results on MedicinalPlant.** All models use ViT-Base/16 at 256×256 . **Bold:** best result. Underline: second-best. †: our method uses *zero* human labels during pre-training.

Method	Pre-train	LP-FT	Val Acc %	Δ Scratch
Scratch (Random)	—	×	49.35	—
ImageNet-21k (Sup.)	21k labels	×	<u>59.99</u>	+10.64
Agri-DINO (Ours)†	SSL	✓	59.67	+10.32

Table 2. **Ablation study on MedicinalPlant.** Starting from a naïve baseline (no LP-FT, 224×224 , no TTA), we progressively add each proposed component. Each row inherits all settings from rows above it. †: improvement over previous row.

Config.	Res.	LP	LLRD	TTA	Acc %	Gain
(a) Naïve	224 ²				35.19	—
(b) + Hi-Res (256 ²)	256 ²				39.82	+4.63
(c) + Lin. Probe warmup	256 ²	✓			52.41	+12.59
(d) + LLRD ($\gamma=0.75$)	256 ²	✓	✓		57.88	+5.47
(e) + TTA (Full)	256 ²	✓	✓	✓	59.67	+1.79

need for large supervised pre-training datasets in the agricultural vision setting.

4.3. Ablation Study

Table 2 isolates the contribution of each proposed component of our fine-tuning protocol. All rows use the same Agri-DINO pre-trained backbone; only the downstream adaptation strategy changes.

The ablation reveals a clear hierarchy of contributions. The **LP warmup stage** (row c) provides the single largest gain (**+12.59 pp**), confirming that catastrophic forgetting of DINO features—caused by unbounded initial head gradients—is the primary failure mode of naive fine-tuning. **High-resolution tokenization** (row b) contributes a consistent **+4.63 pp**, reflecting the importance of fine microstructural cues in botanical discrimination. **LLRD** (row d) adds **+5.47 pp** by calibrating gradient magnitude across transformer depth, protecting low-level texture features in early blocks. **TTA** (row e) provides a modest but consistent **+1.79 pp** at no training cost.

4.4. Convergence Analysis

Table 3 traces validation accuracy and loss across training phases for our full Agri-DINO model. The two-stage transition at epoch 5 (LP $\rightarrow$ LLRD) is clearly visible: validation accuracy climbs sharply as backbone gradients begin propagating, plateauing gracefully without divergence.

Critically, validation loss decreases monotonically and converges smoothly without the divergence or plateau observed in early experiments under naïve fine-tuning (35.19%). The train-val accuracy gap stabilizes at ≈ 25

Table 3. **Training dynamics of Agri-DINO (MedicinalPlant).** Epoch 5 marks the LP $\rightarrow$ LLRD transition. Train/Val Acc are top-1; Val Loss is cross-entropy.

Epoch	Train Acc (%)	Val Acc (%)	Val Loss
1	22.3	24.1	3.401
5	41.8	39.2	2.994
10	62.4	51.6	2.811
20	73.6	56.3	2.694
30	78.8	58.1	2.621
40	82.9	59.4	2.606
50	84.4	59.7	2.598

Table 4. **PlantSeg results under fragmented vs. corrected taxonomy.** Fragmented (300 micro-classes) vs. corrected (20 disease categories). All models use ViT-Base/16 at 256×256 .

Method	Pre-train	Val Acc %	
		Frag. (300 cl.)	Corr. (20 cl.)
Scratch	—	2.42	—
ImageNet-21k	Sup.	2.78	—
Agri-DINO (Ours)	SSL	2.63	—

[†] Corrected results pending relabeling.

pp, consistent with mild generalization on a small (1.3K image) dataset. These dynamics confirm that our LP-FT protocol suppresses overfitting without requiring additional regularization beyond standard dropout and label smoothing.

4.5. Analysis: Micro-Class Fragmentation in PlantSeg

All three initialization methods achieve $\approx 2.6\%$ accuracy on PlantSeg (Table 4), a result that appears to indicate that the dataset is intractable. Our diagnosis reveals a systematic benchmark pathology.

The automated conversion pipeline parsed sample-level file identifiers (e.g., banana-black-sigatoka-10-, banana-black-sigatoka-102-) as distinct taxonomic classes, inflating the class count from the true **20** disease categories to **300 micro-classes**. With 300 classes and only a few hundred total images, most categories contain fewer than 3 training samples—a distribution under which any classifier will approach random chance ($100/300 \approx 0.33\%$). We are actively correcting the pipeline to strip numeric suffixes before taxonomic assignment and will release corrected-taxonomy results in a follow-up evaluation. This pathology is not unique to our dataset: we caution that any pipeline converting YOLO-format annotations with instance IDs to image-classification folders risks the same fragmentation without explicit suffix normalization.

5. Discussion

Self-supervised pre-training as a viable alternative to supervised transfer. The central result of this paper is the near-parity between Agri-DINO (59.67%) and supervised ImageNet-21k (59.99%) on MedicinalPlant, achieved using zero human annotations during pre-training. This finding has a practical implication for agricultural AI: assembling large unlabeled botanical image corpora—from smartphone apps, field cameras, and public image archives—is substantially cheaper than curating expert-annotated datasets. As domain-specific SSL corpora grow (the agricultural community has collectively uploaded tens of millions of images to iNaturalist and PlantNet), we expect the gap between self-supervised and supervised pre-training to further close or reverse.

Why LP-FT matters more for SSL than supervised backbones. Our ablation shows that the LP warmup contributes **+12.59 pp**—the single largest individual gain—yet it is often omitted in agricultural transfer learning papers that exclusively evaluate supervised ImageNet backbones. The reason it matters *more* for SSL is structural: supervised pre-training already aligns features toward discrete classification boundaries, so the classification head gradient is smaller at initialization. DINO features, by contrast, encode view-invariant patch similarities and are *not* pre-aligned to any class taxonomy. A randomly initialized linear head therefore produces far larger initial gradients, necessitating the LP buffer stage.

Limitations and future directions. Three limitations constrain the current work and motivate near-term extensions. First, our evaluation covers only image classification; a complete agricultural foundation model must excel at dense segmentation (lesion mapping), oriented object detection (fruit and weed localization), and regression tasks (crop counting). We will extend Agri-Bench to these modalities using our existing corn-kernel-counting and longitudinal-nutrient-deficiency datasets. Second, all experiments use RGB images; early plant stress is physiologically visible in near-infrared (NIR) and NDVI channels before it is optically apparent in the visible spectrum. We are developing lightweight convolutional spectral band adapters that project an arbitrary number of input channels into the 3-channel input expected by the ViT patch embedding, enabling multi-spectral fine-tuning without architectural modification to the backbone. Third, scaling input resolution beyond 256×256 toward 448×448 incurs quadratic attention cost. We will explore attention-based background token pruning—discarding uninformative soil and sky tokens at the beginning of each block—to maintain accuracy gains at manageable computational cost.

6. Conclusion

We presented Agri-DINO, a self-supervised Vision Transformer foundation model for fine-grained agricultural and botanical vision. By pre-training on a curated 5.17M-image unlabeled corpus and adapting downstream tasks via our Two-Stage LP-FT protocol with domain-calibrated LLRD, Agri-DINO achieves a **+10.32 pp** absolute accuracy gain over from-scratch ViT training and matches supervised ImageNet-21k performance— all without human annotation during pre-training. A systematic ablation isolates the contribution of each protocol component, and our analysis of micro-class fragmentation in PlantSeg establishes a practical pitfall for the broader agricultural AI community. We will release our pre-trained checkpoints, WebDataset shards, and Agri-Bench evaluation scripts to facilitate reproducible research in agricultural computer vision.

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